

X · Ch. None

leph202 — extracted questions & answers

8 top-level · 18 logical questions · 4 with an answer in the document · 10 explicitly not in the document

10.1. Example 10.1

p. 8 · unknown · worked_example · blocks: blk_000087

a. (a) When monochromatic light is incident on a surface separating two media, the reflected and refracted light both have the same frequency as the incident frequency. Explain why?

p. 8 · conceptual · worked_example · blocks: blk_000088

Answer: (a) Reflection and refraction arise through interaction of incident light with the atomic constituents of matter. Atoms may be viewed as oscillators, which take up the frequency of the external agency (light) causing forced oscillations. The frequency of light emitted by a charged oscillator equals its frequency of oscillation. Thus, the frequency of scattered light equals the fr

source blocks: blk_000092

b. (b) When light travels from a rarer to a denser medium, the speed decreases. Does the reduction in speed imply a reduction in the energy carried by the light wave?

p. 8 · conceptual · worked_example · blocks: blk_000089

Answer: (b) No. Energy carried by a wave depends on the amplitude of the wave, not on the speed of wave propagation.

source blocks: blk_000093

c. (c) In the wave picture of light, intensity of light is determined by the square of the amplitude of the wave. What determines the intensity of light in the photon picture of light.

p. 8 · conceptual · worked_example · blocks: blk_000090

Answer: (c) For a given frequency, intensity of light in the photon picture is determined by the number of photons crossing an unit area per unit time.

source blocks: blk_000094

10.2. Example 10.2 Discuss the intensity of transmitted light when a polaroid sheet is rotated between two crossed polaroids?

p. 17 · conceptual · worked_example · blocks: blk_000261

Answer (partial): Solution Let I_0 the first polariser P 1 second polariser P will be be the intensity of polarised light after passing through . Then the intensity of light after passing through 2 where θ is the angle between pass axes of P 1 and P 2 . Since P 1 and P 3 are crossed the angle between the pass axes of P 2 and P 3 will be $(\pi/2\theta)$. Hence the intensity of light emerging from P 3 will be

source blocks: blk_000262, blk_000263, blk_000264, blk_000265

10.1. 10.1 Monochromatic light of wavelength 589 nm is incident from air on a water surface. What are the wavelength, frequency and speed of (a) reflected, and (b) refracted light? Refractive index of water is 1.33.

p. 19 · conceptual · solved_question · blocks: blk_000286

NOT IN DOCUMENT — no answer present in the source

10.2. 10.2 What is the shape of the wavefront in each of the following cases:

p. 19 · conceptual · solved_question · blocks: blk_000287

a. (a) Light diverging from a point source.

p. 19 · conceptual · solved_question · blocks: blk_000288

NOT IN DOCUMENT — no answer present in the source

b. (b) Light emerging out of a convex lens when a point source is placed at its focus.

p. 19 · conceptual · solved_question · blocks: blk_000289

NOT IN DOCUMENT — no answer present in the source

c. (c) The portion of the wavefront of light from a distant star intercepted by the Earth.

p. 19 · conceptual · solved_question · blocks: blk_000290

NOT IN DOCUMENT — no answer present in the source

10.3. 10.3

p. 19 · unknown · solved_question · blocks: blk_000291

a. (a) The refractive index of glass is 1.5. What is the speed of light in glass? (Speed of light in vacuum is $3.0 \times 10^8 \text{ m s}^{-1}$)

p. 19 · numerical · solved_question · blocks: blk_000291

NOT IN DOCUMENT — no answer present in the source

b. (b) Is the speed of light in glass independent of the colour of light? If not, which of the two colours red and violet travels slower in a glass prism?

p. 19 · conceptual · solved_question · blocks: blk_000292

NOT IN DOCUMENT — no answer present in the source

10.4. 10.4 In a Young's double-slit experiment, the slits are separated by 0.28 mm and the screen is placed 1.4 m away. The distance between the central bright fringe and the fourth bright fringe is measured to be 1.2 cm. Determine the wavelength of light used in the experiment.

p. 19 · numerical · solved_question · blocks: blk_000293

NOT IN DOCUMENT — no answer present in the source

10.5. 10.5 In Young's double-slit experiment using monochromatic light of wavelength λ , the intensity of light at a point on the screen where path difference is λ , is K units. What is the intensity of light at a point where path difference is $\lambda/3$?

p. 19 · numerical · solved_question · blocks: blk_000294

NOT IN DOCUMENT — no answer present in the source

10.6. 10.6 A beam of light consisting of two wavelengths, 650 nm and 520 nm, is used to obtain interference fringes in a Young's double-slit experiment.

p. 19 · numerical · solved_question · blocks: blk_000295

a. (a) Find the distance of the third bright fringe on the screen from the central maximum for wavelength 650 nm.

p. 19 · numerical · solved_question · blocks: blk_000296

NOT IN DOCUMENT — no answer present in the source

b. (b) What is the least distance from the central maximum where the bright fringes due to both the wavelengths coincide?

p. 19 · conceptual · solved_question · blocks: blk_000297

NOT IN DOCUMENT — no answer present in the source